Activity 3.2: Triangle Properties

A

Name: _____ Partner's name: _____ Date: _____

Directions: Find the value or answer asked for. Show your work. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

$AB = AC$ in isosceles $\triangle ABC$. $m\angle B = (2x + 15)°$ and $m\angle C = (4x - 5)°$. Find x.

Answer: _____ Match? ☐ Yes ☐ No

2

A triangle is equilateral with sides $3x - 3$, $2x + 3$, and $x + 9$. Find its perimeter.

Answer: _____ Match? ☐ Yes ☐ No

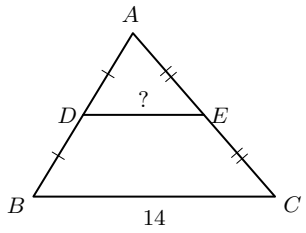
3

Two sides of a triangle measure 4 and 9. What is the smallest possible whole-number length of the third side?

Answer: _____ Match? ☐ Yes ☐ No

4

D and E are midpoints of $\overline{AB}$ and $\overline{AC}$. Find DE.



Answer: _____ Match? ☐ Yes ☐ No

5

P is on the perpendicular bisector of $\overline{AB}$. $PA = 3x + 1$ and $PB = 5x - 7$. Find PA.

Answer: _____ Match? ☐ Yes ☐ No

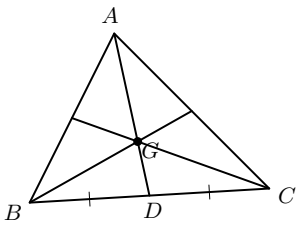
6

P is on the bisector of $\angle ABC$. P is $2x + 3$ from $\overline{BA}$ and $4x - 5$ from $\overline{BC}$. Find the distance from P to $\overline{BA}$.

Answer: _____ Match? ☐ Yes ☐ No

7

G is the centroid of $\triangle ABC$ and median $AD = 18$. Find AG.



Answer: _____ Match? ☐ Yes ☐ No

8

Classify $\triangle ABC$ by its sides and by its angles: $AB = 6$, $BC = 8$, $CA = 10$.

Answer: _____ Match? ☐ Yes ☐ No

Activity 3.2: Triangle Properties

B

Name: _____ Partner's name: _____ Date: _____

Directions: Find the value or answer asked for. Show your work. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

$PQ = PR$ in isosceles $\triangle PQR$. $m\angle Q = (3x + 5)°$ and $m\angle R = (5x - 15)°$. Find x.

Answer: _____ Match? ☐ Yes ☐ No

2

A triangle is equilateral with sides $4x - 9$, $x + 9$, and $5x - 15$. Find its perimeter.

Answer: _____ Match? ☐ Yes ☐ No

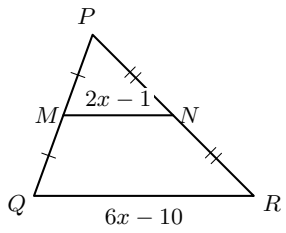
3

Two sides of a triangle measure 5 and 10. What is the smallest possible whole-number length of the third side?

Answer: _____ Match? ☐ Yes ☐ No

4

M and N are midpoints of $\overline{PQ}$ and $\overline{PR}$. $MN = 2x - 1$ and $QR = 6x - 10$. Find MN.



Answer: _____ Match? ☐ Yes ☐ No

5

Q is on the perpendicular bisector of $\overline{CD}$. $QC = 2x + 5$ and $QD = 4x - 3$. Find QC.

Answer: _____ Match? ☐ Yes ☐ No

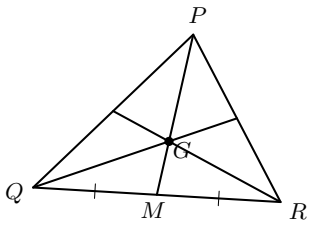
6

Q is on the bisector of $\angle XYZ$. Q is $3x - 1$ from $\overline{YX}$ and $x + 7$ from $\overline{YZ}$. Find the distance from Q to $\overline{YX}$.

Answer: _____ Match? ☐ Yes ☐ No

7

G is the centroid of $\triangle PQR$. The median from P meets $\overline{QR}$ at M, and $GM = 6$. Find PG.



Answer: _____ Match? ☐ Yes ☐ No

8

Classify $\triangle PQR$ by its sides and by its angles: $PQ = 5$, $QR = 12$, $PR = 13$.

Answer: _____ Match? ☐ Yes ☐ No